Seeyan Newaz

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Education

The University of Texas at Dallas — **Aug 2022 – May 2026**
Bachelor of Science in Computer Science — *GPA: 3.92*

- **Relevant Coursework:** Computer Architecture, Discrete Maths, Data Structures & Algorithms, Software Engineering, Programming Languages & Paradigms, Probability & Statistics, Database Systems, Linear Algebra, Computer Networks, Digital Logic, Automata Theory
- **Organizations:** Association for Computing Machinery at UTD (Member), Artificial Intelligence Society at UTD (Member), Bangladeshi Student Organization at UTD (Member)
- **Achievements:** Academic Excellence Scholarship, Texas Competitive Scholarship, Dean's List, Edexcel High Achievers' Award, The Daily Star Award

Technical Skills

Languages: Python, Java, C/C++, C#, HTML/CSS, JavaScript/TypeScript, Bash, SQL, PHP, Dart, Assembly

Frameworks/Libraries: Spring Boot, React.js, Next.js, Angular, Node.js, .NET, Flask, Flutter, Tailwind CSS, Pandas, NumPy, scikit-learn, SciPy, MNE, D3.js, Chart.js

Backend & Data: PostgreSQL, Redis, Redis Streams, MySQL, Firebase, Oracle, Spring Data JPA, Hibernate, SQLAlchemy, Flyway

Developer Tools & Cloud: Git, Docker, Docker Compose, Kubernetes, Maven, Linux, Postman, JIRA, AWS, Microsoft Azure, OpenAI API, Prometheus, Micrometer

Experience

Machine Learning Engineer — **Jan 2026 – May 2026**
The University of Texas at Dallas — *Richardson, TX*

- Built a reproducible **Python EEG classification pipeline** using **MNE**, **SciPy**, **pandas**, & **scikit-learn** for Alzheimer's detection.
- Implemented **Welch PSD** feature extraction across **19** EEG channels, deriving Delta, Theta, Alpha, Beta, & ratio-based biomarkers for downstream classifiers.
- Engineered posterior-channel, log-transformed, & derivatives-based features, improving **Random Forest accuracy** to **80.0%** & **ROC-AUC** to **0.861**.
- Validated **Logistic Regression** & **Random Forest** with subject-level **5-fold CV**, preventing data leakage & improving reliability of participant-level evaluation.

Software Developer Intern — **May 2025 – Aug 2025**
Paycom — *Irving, TX*

- Engineered a full-stack **C#/.NET** PDF data extraction pipeline for new client onboarding, projected to generate **$100K+** savings.
- Cut manual setup by up to **5** days per client by building a **React**-based annotation tool & intelligent table-detection algorithms.
- Reduced manual labeling/review time by **70%** by developing custom **NLP/NER** models for structured entity extraction.
- Designed inference & auto-classification backend services, enabling automated data mapping & speeding up downstream analytics.

Backend Software Developer — **Feb 2024 – Apr 2024**
Association for Computing Machinery – The University of Texas at Dallas — *Richardson, TX*

- Built a **Dart/Flutter** events app for **30K+** students across **50+** university organizations by working in a **Scrum-based 5**-person team.
- Delivered **90%** accurate personalized event recommendations by integrating a **GPT-4 Turbo** chatbot into the user experience.
- Enhanced data retrieval efficiency by **50%** by designing a **Firestore** schema for real-time synchronization.

Peer Tutor for Calculus & Linear Algebra — **Aug 2023 – May 2026**
Student Success Center – The University of Texas at Dallas — *Richardson, TX*

- Achieved **60%** average grade improvement over **500+** undergraduates by delivering personalized **1:1** tutoring.
- Enhanced concept mastery & confidence of students by tailoring explanations & practice to diverse learning styles.
- Boosted exam readiness & achieved **100%** student satisfaction by creating custom problem sets & step-by-step solutions.

Projects

FlowForge | ***Java, Spring Boot, PostgreSQL, Redis Streams, Docker, Micrometer*** — **Jun 2026**

- Built a **distributed, fault-tolerant job engine** for durable async execution using **Spring Boot REST APIs, Redis Streams, PostgreSQL, & Docker**.
- Increased failure resilience with **2s–60s exponential backoff** by implementing retry scheduling, **attempt-level audit logs**, & **dead-state handling**.
- Eliminated **Redis/PostgreSQL consistency races** during dispatch by using **post-commit publishing** for durable job consumption.
- Added production-grade **observability & recovery** with **5s worker heartbeats**, **30s stale-RUNNING detection**, lock-ownership checks, & **Prometheus metrics**.

Trace | ***Python, Pandas, NumPy, scikit-learn, Matplotlib, yfinance, D3.js, Chart.js*** — **Apr 2026**

- Built a **post-earnings stock prediction platform** at FinHack 2026 for **64** earnings events across **8** AI-linked companies using market & text signals.
- Created company, peer, regulation, & event-level sentiment features from **news, transcripts, press releases, & video data**.
- Engineered a leakage-safe **NLP pipeline** with **7**-day windows, spillover features, AI keyword counts, & conflict indicators.
- Raised test **accuracy** from **50.0%** to **62.5%** & **ROC-AUC** from **0.667** to **0.792** using chronological model evaluation.
- Built an interactive **D3.js/Chart.js** dashboard with model comparison, contagion mapping, case studies, & feature-importance views.

LedgerLens | ***Docker, Spring Boot, Angular, PostgreSQL, OpenAI API*** — **Feb 2026**

- Developed an **AI-driven, full-stack** web app using **GPT-5.2** that enables users to upload receipt images & track expenditures.
- Cut manual receipt processing by **50%** through automated extraction & analysis of expenditure data.
- Built a dynamic **Angular** frontend that displays all-time/monthly insights, real-time data visualization & spending trends.
- Implemented a scalable backend with **Spring Boot, PostgreSQL,** & **Docker** for persistent storage & consistent deployment.

InboxIntel | ***Python, Streamlit, PostgreSQL, Gmail API, OpenAI API*** — **Nov 2025**

- Engineered an **AI-powered, full-stack** intelligent inbox assistant using **GPT-5.1** to retrieve & process unread messages in real time.
- Cut manual email processing time by **60%** through automated summaries, prioritization, task extraction, & reply drafts.
- Designed a responsive **Streamlit** interface by building dynamic filters & metrics.
- Reduced duplicate processing by implementing **PostgreSQL** state management & deduplicated query flows.